Long Form Project - Game Design Documentation

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Assignment: LFP Milestone 2

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* Red stars indicate work that has been newly added or work that has been changed for this submission.

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Game Research

Brief synopsis of Our Game

Our game is a narrative-driven puzzle game wherein the player must overcome a virus *that has corrupted their code. The narrative follows the player, who takes on the role of a game developer, playtesting their own game. This game features a tile based grid that players traverse to reach an end point. This end point will be blocked by obstacles that can be overcome through the mechanical switching *of different map modes. You switch from the 'normal mode' map to the 'safe mode' map wherein the tile layout is similar, but *obstacle properties have changed. Movement made on the safe map is reflected on the normal map, and vice versa. *Some player mechanics are only available in specific map modes.
As the player moves through the map levels, they encounter The Virus. The Virus is the antagonist of the narrative and will 'corrupt' the map that the player is on. Corruption occurs through the changing of pathways, obstacles, map layouts, and mechanics. *These changes are movement and puzzle based, with players needing to retrace steps and drop and pick up items, all whilst switching between map modes. The player must try to destroy The Virus by completing the puzzles of the game and using the normal-to-safe switching to clear/navigate around all Virus changes .

Cloned Games

The games, and their relevant mechanics, that our group took inspiration from are as follows:

- **The Silent Age - Time Travel Mechanic**

The Silent Age is a point-and-click mystery adventure game. Players take on the role of a janitor working at a secretive research corporation. The player discovers an extinction level threat within the corporation labs and is given a time machine to prevent this threat. Players use this time machine to toggle/move between the present (1972) and the post-apocalyptic future (2012) to solve logical inventory and obstacle puzzles in an attempt to change the future and prevent humanity's demise.
The time travel mechanic operates with the click of the time-machine and effectively puts the player in the exact position in the future as they would be in the past, and vice versa. Use of the time machine is finite and can only be recharged when the player exposes the time machine to sunlight (charge-dependent).

How we will use this mechanic: The charge-based time travel between two of the same points in time will be taken from *The Silent Age* and adapted for our own game. We will create a map switching mechanic wherein the player can switch from a 'normal' map to a 'safe' map. The player's location on the maps will be identical when switched, similar to what is seen in *The*

Silent Age. Normal and safe maps will be similar in layout, but not identical. Any changes made on the safe map will reflect on the normal map.
Additionally, map switching will be accompanied by a visual change. For example, in *The Silent Age* when players go from past to future, the architecture of the location won't change, but the visuals will. See Figure 1 below. We will similarly keep the base layout of maps the same, but we will go from a modern style on the normal map to a *pixel art style on the safe map.

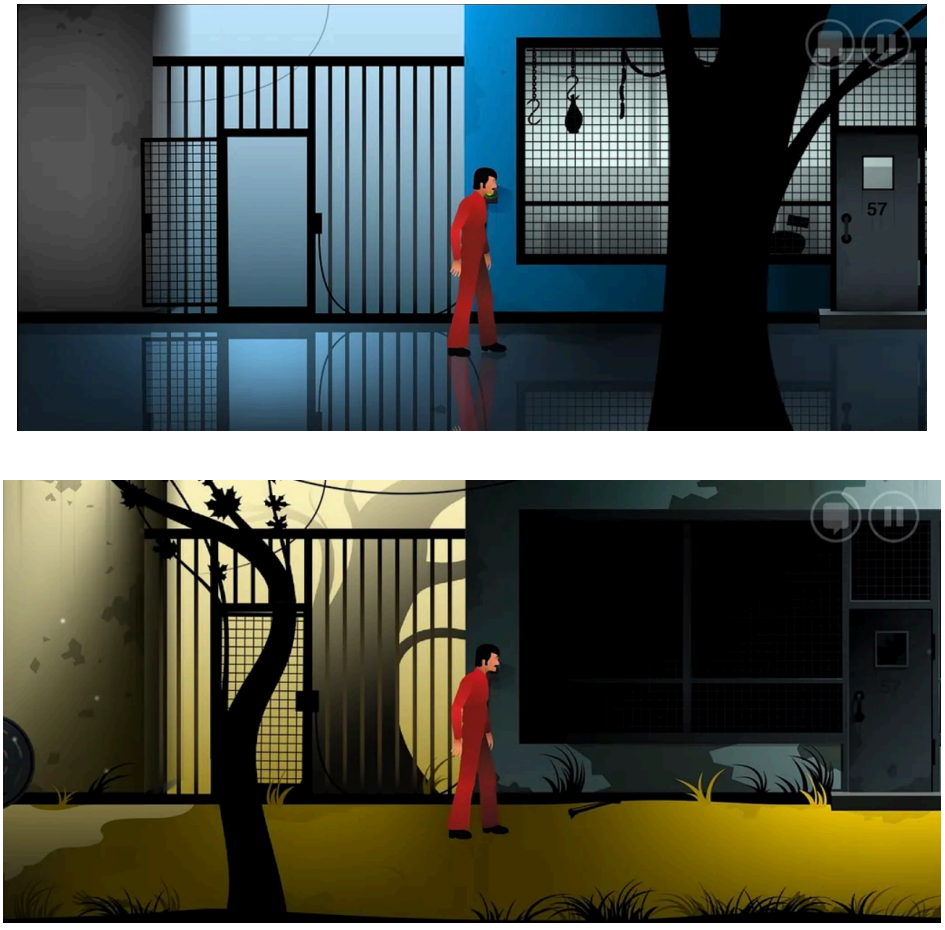


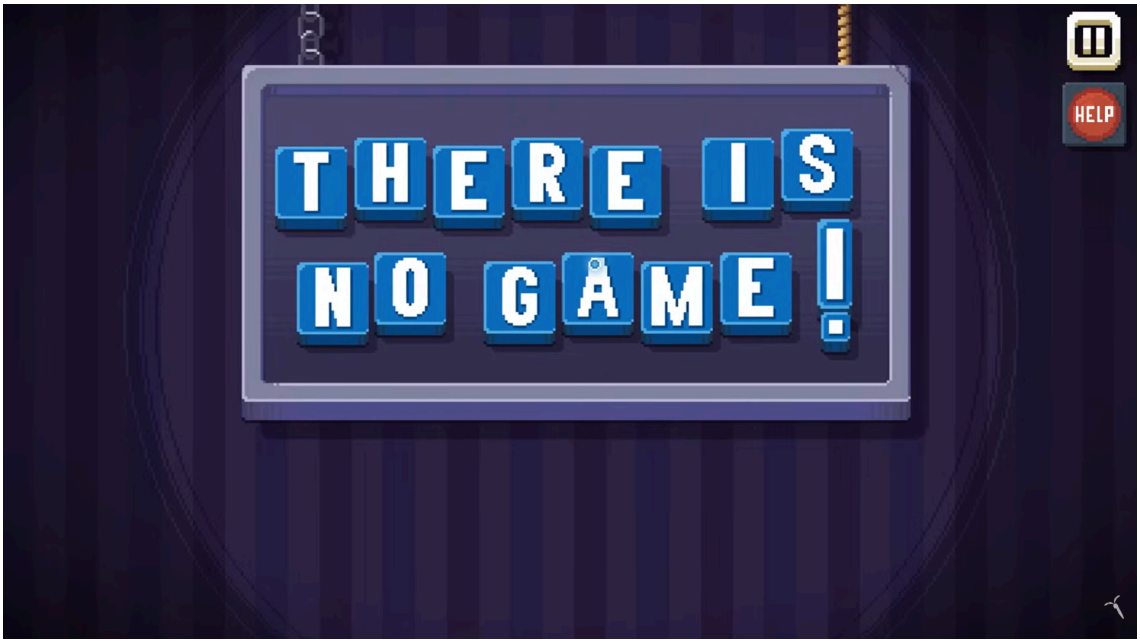
Figure 1: Still images from The Silent Age *game (Allred 2012). The top screenshot features a prison in present/past time (1972). The bottom screenshot features the same prison in the future (2012). This is an example of the visual changes made to the same location when using the time machine mechanic.*

- **There is No Game - Antagonistic Narrator**

There is No Game: Wrong Dimension is a meta, point-and-click comedy adventure game which heavily breaks the fourth wall. This is done through a title screen and sentient game narrator who repeatedly insists that there is no game and begs the player to stop playing and leave. Progress in the game involves ignoring the narrator, breaking UI and interacting with the environment. The entire experience heavily features the tension between the player taking

action and the protesting/antagonistic narrator making their own actions and reacting to the player's actions.

How we will use this mechanic: We will be utilising the antagonistic narrator who acts as an active system that tries to hinder the player from progressing through the game. Our game will adapt the narrator's role by transforming them into The Virus, a system and narrator who actively changes the player's game by manipulating the tile-based puzzles and environment that the player is trying to solve. This will mirror *There is No Game*'s narrator by having the Virus speak to the player. The player will *respond to the virus, and overcome map changes The Virus makes, causing The Virus to *further 'act out'. This "acting out" will be in the form of the Virus adding extra obstacles to the puzzle, changing puzzle solutions and constantly forcing the player to think about how the map switch mechanic can manipulate the Virus to their advantage to solve the puzzle.



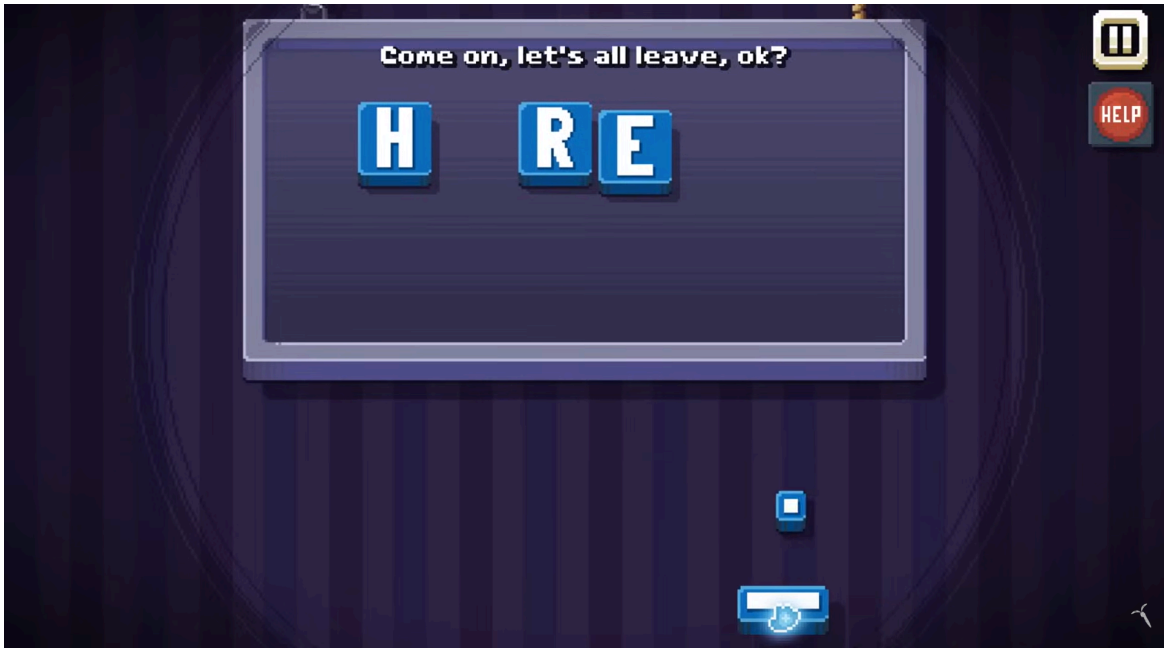


Figure 2: Still images from There is no Game: Wrong Dimension. *The first image shows the title screen which the narrator says cannot be interacted with. The second image shows the player, ignoring the narrator and interacting with the game title letters which turns the "solution" of the puzzle into a game of Brick Breaker.*

Category and Subcategory Analysis

The games used as inspiration for this project primarily fall within the **adventure** genre, with subtle **puzzle** genre elements . While both *The Silent Age* and *There Is No Game: Wrong Dimension* are classified primarily as adventure games, they use puzzle-solving as an important method of progressing through their narratives. The game being developed takes inspiration from both of these games but places a greater emphasis on puzzle-solving as its primary form of gameplay through the tile-based environment.

The Silent Age

Genre: Mystery Adventure
Sub-Genre/Elements: Point-and-Click Puzzle Adventure

The Silent Age can be classified primarily as an **adventure game**, as its gameplay focuses on exploration, interacting with the environment, discovering information and progressing through a

narrative. The player controls the protagonist through different environments and must interact with objects and characters to uncover the story. The game can further be described as a **point-and-click puzzle adventure**, as progression frequently requires the player to identify relationships between objects, environments and events in order to overcome obstacles.

The game's time-travel mechanic is particularly important to its puzzle elements. The player can switch between two versions of the same environment, with differences between the two time periods allowing the player to solve environmental puzzles. Therefore, the adventure genre provides the narrative and exploration framework, while the puzzle elements provide the primary method of interacting with and progressing through the environment.

There Is No Game: Wrong Dimension

Genre: Comedy Adventure
Sub-Genre/Elements: Point-and-Click Puzzle, Meta-Fiction

There Is No Game: Wrong Dimension can also be classified as a "**comedic" adventure game**, as the player progresses through a series of environments by interacting with objects, solving problems and uncovering the game's narrative. Its gameplay is heavily based around **puzzle-solving**, with the player required to experiment with the environment and identify unconventional solutions to progress.

The game also contains strong **meta-fictional** elements, as it directly acknowledges the player's presence and breaks the conventional boundary between the player, narrator and the game. The narrator is aware of the player's actions and actively attempts to prevent them from progressing (see figure 2). This relationship between the player, the game and its narrator forms an important part of both the game's narrative and gameplay.

Our Game

Genre: Puzzle Adventure
Sub-Genre/Elements: 2D Topdown, Narrative Puzzle, **Tile-Based Puzzle**, Meta-Fiction

Our game primarily falls within the **puzzle genre**, as its main gameplay objective is to solve environmental puzzles in order to progress through the game. The player navigates a tile-based grid and must use the relationship between the **Normal mode** Map and **Safe mode** Map to overcome obstacles and reach the endpoint of each level. The map-switching mechanic therefore forms the central puzzle-solving system rather than simply being an additional feature, while the tile-based form of the game environment forms the actual "puzzle space".

The game also contains elements of the **adventure genre**, particularly through its narrative progression and exploration of different game environments. However, unlike *The Silent Age* and *There Is No Game*, where puzzle-solving is contained within a broader point-and-click

adventure structure, our game uses puzzle-solving and tile-based movement as its primary gameplay.

The game also incorporates **meta-fictional** elements inspired by *There Is No Game: Wrong Dimension*. The Virus is aware of the player's actions (only within Normal Mode) and actively interferes with the game world by changing pathways, obstacles, layouts and mechanics. This creates a relationship between the player and the game system similar to the antagonistic relationship between the player and narrator in *There Is No Game*.

The game's genre is therefore derived from the combination of the genres and elements found in the cloned games. The adventure elements are informed by both *The Silent Age* and *There Is No Game*, while the game's puzzle focus is influenced by the environmental and logical puzzles found in both titles. The tile-based puzzle structure and map-switching system adapt the two-world puzzle concept of *The Silent Age* into a different gameplay format, while the meta-fictional Virus adapts the self-aware and antagonistic game system of *There Is No Game*. These elements are combined to create a more puzzle-focused adventure game with its own distinct gameplay and narrative identity.

Game Interrogation

Core Game Question

The core idea behind the development of this game is investigating how actively subverting player expectations and understanding of game mechanics and solutions affect a player's gameplay engagement and overall decision-making within a game.

Hypothesis

To what extent does an antagonistic narrative game element, The Virus, establish player expectations about how game mechanics and puzzles function, and how does its deliberate subversion of these expectations affect the player's experience and decision-making?

Detailed Hypothesis Explanation

* The game mechanics we refer to in our hypothesis are the systems and actions available to the player that allow * interaction and progression through the game. These include player movement, *puzzle and object interaction alongside the switching between map modes (normal and safe). As players progress, they learn how these mechanics work and begin to form expectations about how the game will respond to their actions. For example, one of the game's puzzles is built around a "pressure plate" mechanic, where the player has to drop an item on

this pressure plate for a door/gate to open. * The player then forms an expectation that a specific pressure plate will open a certain door/gate.

* In our game, we then subvert these expectations by the introduction of The Virus. The Virus is an antagonistic narrative game element that interacts directly with the player and actively interferes with their progression. While The Virus functions as an antagonist within the narrative, it also directly affects gameplay by changing the environment, obstacles, pathways and altering mechanics that the player uses to solve puzzles. This deliberate interference with the systems already learnt means that strategies which were previously successful may no longer work. * In doing so, we force the player to reconsider their understanding of the game's systems and promote adaptation to their approach towards puzzle solving. Using the "pressure plate" example; the player will have the idea that, for example, "Pressure plate 1 will open gate 2". The Virus will then deliberately switch pressure plate positions and alter the puzzle solution completely. The player will need to refactor what they know about the puzzle, hence we have subverted their expectations. * The player must now use new mechanics of switching between normal and safe mode to proceed forward and solve the puzzle.

The purpose of this project is therefore to investigate how players respond when their established expectations of game mechanics and puzzle-solving are deliberately disrupted/or changed. In particular, we are interested in whether this subversion changes the player's decision-making, encourages experimentation and adaptation, and affects their overall experience of the game. *However, we are still in the process of investigating this player response and have not drawn any clear conclusion. As of yet, we are actively playtesting and recording player feedback through Google Forms. This is helping us to establish the thinking behind our players actions, which shapes the game's iterations. Only after the game has been fully developed can we answer our hypothesis.

During playtesting, players will be observed as they encounter mechanics that initially behave consistently and establish expectations, followed by puzzles where The Virus deliberately subverts those expectations. Their reactions, problem-solving behaviour and decision-making can then be recorded through observation, questionnaires or interviews. Comparing player responses before and after their expectations are subverted can provide evidence of how this design approach affects the player's experience and approach to solving puzzles.

Design Goals

1. **Establish a Reliable Gameplay Foundation**

Create a gameplay loop in which the player develops a baseline understanding of the game's mechanics, conventions and systems before The Virus begins to subvert their expectations. Core rules that are necessary for understanding and navigating the game should remain consistent, allowing the player to distinguish between the game's established rules and deliberate interference caused by The Virus. This foundation should prevent the player from feeling that the game is arbitrary, unfair or excessively frustrating/confusing when mechanics are subverted.

More specifically,the parts of the gameplay foundation that need to remain reliable and consistent should be the learned puzzle mechanics themselves. For example, if a player has learned that doors are opened by finding and using a key, then the virus should never disrupt that relationship. The Virus should only disrupt ways in which that relationship/interaction happens. For example, the Virus can make it so that doors can only be opened in safe mode, while the key will be returned to its original position if the player switches back to normal mode while holding said key in their inventory- but the key-to-door relationship should still remain intact.

2. **Deliberately Subvert Player Expectations**

Create narrative beats, Virus mechanics and puzzle sections that intentionally challenge expectations that players have developed through previous interactions with the game. Once a mechanic or puzzle convention has been established, The Virus should be able to interfere with it through intentionally confusing but logical alterations. These changes should be intentional and recognisable as the actions of The Virus rather than appearing to be inconsistent/glitchy game design.

This should be done in two ways: the first is in clear communication design. The player should be able to see *dialogue text with the Virus, a particle effect, sound effect or other visual communication to indicate The Virus at play. This plays an important role in allowing players to see the intentionality of The Virus's actions. So as previously mentioned, the gameplay should not look glitchy and bugged. The player should rather experience a "glitchy" feel from the Virus' actions which they should fully know is intentionally designed from the virus.

The second way to achieve this is through deliberate narrative beats. For example, if a player has opened a gate using a specific pressure plate, the player should see that the gate is opened. When the player tries to enter the gate, the gate should close on them with Virus text hinting that the pressure plates are switched. This will effectively communicate how the puzzle was changed, which will reduce player frustration because they will actively be able to plan how to resolve the puzzle with this Virus communication.

3. **Encourage Player Adaptation and Experimentation**

Design puzzles that require players to reconsider their existing understanding of the game's mechanics when The Virus interferes with them. Subversion should encourage players to experiment with different approaches, test the boundaries of the game's systems and adapt their strategies rather than simply preventing progress. The player's ability to learn, adapt and develop new solutions should therefore be central to progression.

Adaptation and experimentation come from designing puzzles that create a player expectation, then creating a Virus puzzle change which causes the player to question that expectation. The player should know enough about the puzzle that they wish to investigate what caused the change, rather than being frustrated that a change happened because they do not know what happened. Using the earlier pressure plate example, the investigation and experimentation happens when the player goes back and investigates whether the pressure plate they expected to open a certain gate still opens that same gate.

4. **Seamlessly Integrate Narrative and Gameplay**

Ensure that The Virus's role as a narrative antagonist is directly connected to its gameplay actions. The Virus should communicate with the player and establish its personality through dialogue and narrative interactions while simultaneously affecting the puzzles and game world. This connection should make the Virus's interference feel like an intentional part of the narrative rather than a separate gameplay mechanic.

This should be done through effective Virus dialogue and gameplay interactions.

5. **Create Meaningful Player Reactions**

Design the Virus's interventions to produce observable changes in how players approach the game. The goal is for players to experience reactions such as surprise, uncertainty, allowable frustration, curiosity or increased experimentation when their expectations are subverted, while still feeling that the puzzles are fair and solvable. These reactions will allow the effectiveness of the game's central design concept to be evaluated through playtesting.

Core Gameplay Loop

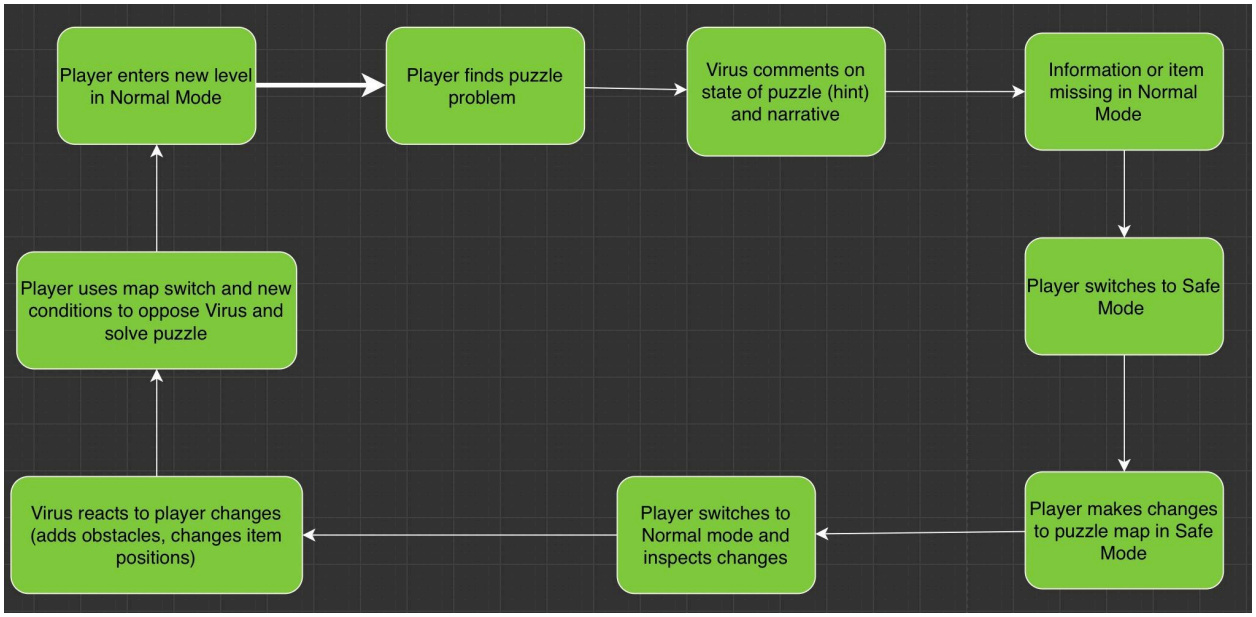


Figure 3: Core gameplay loop for each level.

Systems and Mechanics

Mode Switching (Time Travel mechanic)

The game will be played in two different modes, safe mode and normal mode. In normal mode the player will have to complete puzzles, while the player can get parts of the puzzle solution from safe mode, but can't complete the puzzle in safe mode.
There is an antagonist to the player, called The Virus, in normal mode which **cannot affect the player in safe mode**. This antagonistic Virus will eventually 'corrupt' normal mode by changing puzzles on the map whilst the player is playing and engaging with the world.

The purpose of this mode switching is to create a cause and effect relationship between safe and normal mode. The player needs to take actions or movements that they can **ONLY** perform in one mode to create a solution within the other mode. The successful interaction between these 2 modes is the ultimate solution to being able to solve obstacles and problems that The Virus causes.

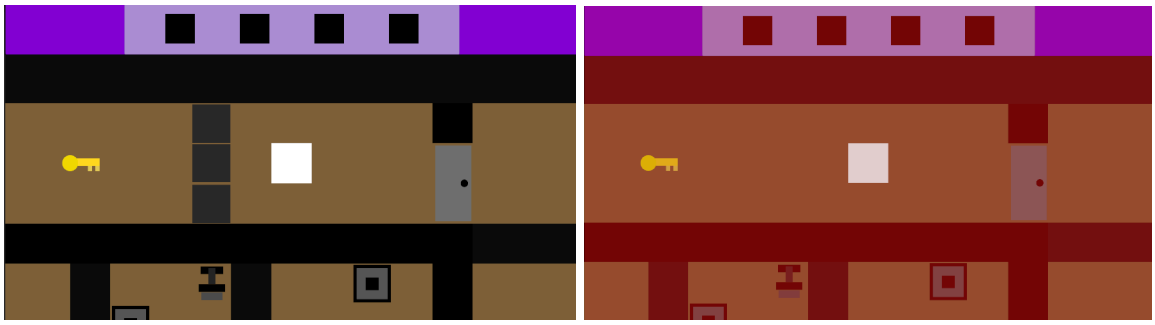


Figure 4: Normal Mode Virus Interference (left) vs Safe Mode Puzzle Solution to that interference (right).

Corruption

In the normal mode, the player will be able to complete the puzzle, but they are constantly getting opposed by The Virus. * This opposition is present in the form of corrupted items. Corrupted items work similarly to normal items, but there are a few key differences. Corrupted items are visually different to their normal counterparts and cannot be picked up in normal mode. These objects can only be interacted with in safe mode, and often require a specific inventory slot to be available in order to pick them up.

Safe Mode

In safe mode the player will not be able to complete the puzzle, but they will find some of the puzzle's solutions in safe mode that were corrupted by The Virus. * For example, doors and pressure plates cannot be interacted with in safe mode. However, the key or statue needed to unlock those puzzles will be found, and accessible, in safe mode. This forces the player to switch to safe mode to progress.
* By limiting the player mechanics in safe mode, the player cannot beat the level entirely in only one mode. This incentivises players to switch between modes constantly. This creates a heightened sense of interaction and adds new depth to possible difficulty in puzzles. This is due to safe mode being one "part"of the puzzle solution, and not the whole solution itself. If these mechanical limitations were not present, the player would always stay in safe mode because this is a map without any Virus problems, making it easier to complete.

***Note to group for next submission**
Consider implementing a mechanic that keeps the player in one mode until they find something to switch to the other mode. If the player can always switch between modes then the tension and relationship between the 2 modes will feel less engaging as the player will have an instant solution to every problem if the player recognizes which mode to be in at a specific point in time. This is an issue in our current build of the game *

The Virus

The virus is the main antagonist stopping the player from progressing and beating the game. The virus can only affect the player and the game in normal mode, which makes safe mode *a place for the player to avoid the Virus's interference. The Virus's system is communicated to the play in a narrative and visual style. Narratively, dialogue will pop up on the bottom of the screen. Visually, this dialogue is supplemented by an icon (see bottom left of figure 4) to represent that the Virus is talking. Visual changes also appear on the map when the Virus system is activated. For example, when The Virus moves a key, the player can actively see the key icon disappear from one spot and appear at the next.

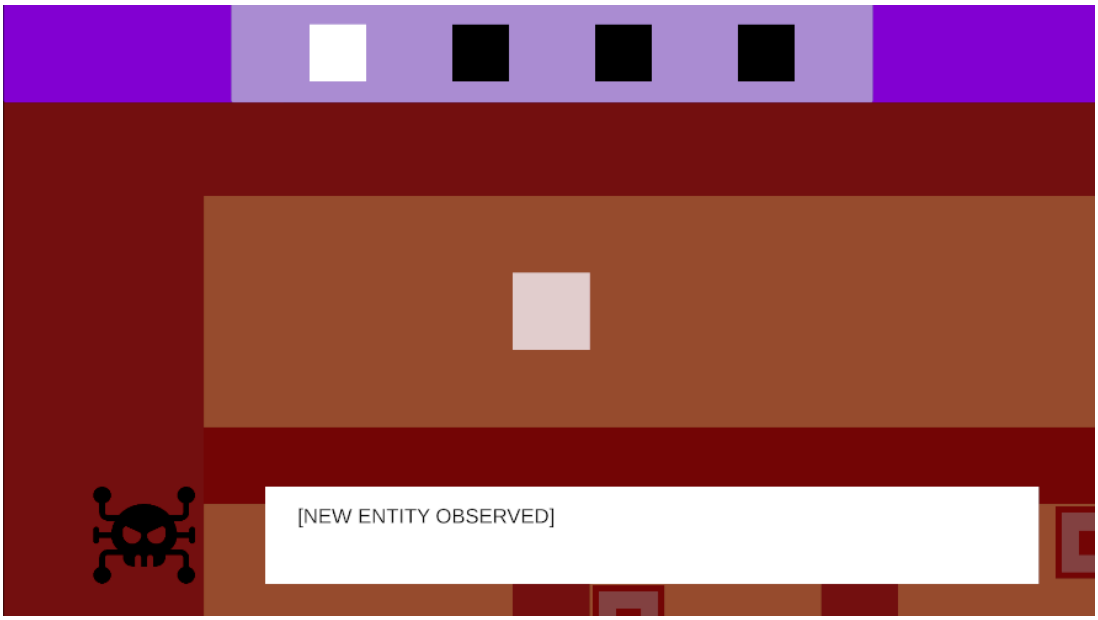


Figure 5: Narrative and visual communication of The Virus

The Virus has different ways of opposing the player which can be seen in the following list:

- Removal of mechanics
    - The Virus can remove a mechanic from the player preventing them from using that mechanic.
- Adding of obstacles
    - The Virus can corrupt game elements to make it impossible for the player to pass in normal mode. Specifically, the Virus can create blocked cells which the player cannot move past. Because the Virus does not exist in Safe Mode, these cells are not blocked anymore and thus can be traversed through.
- Setting Traps
    - The Virus can set traps on specific tiles to cause a game over for the player.
- Changing Element Positions
    - The Virus can change the position of different game elements.
- Corrupting information
    - The Virus can corrupt the information of the player in order to hinder their progress.

Interaction System

* Multiple objects are present throughout every level that require interaction. Most interactions occur with the ‘E’ key or the ‘C’ key. These interactions include: picking up items, dropping items, opening locks, scanning cards. Certain interactions require objects in the inventory to be selected, which is done with the left mouse button. For example, locked safes can only be unlocked when the player left clicks the code object and then presses E.

* Inventory System

The inventory is always present as a bar on the top of the screen. There are 4 inventory slots available for the player to fill. These slots can be filled with items found on the map by moving the player icon over the item and then pressing the ‘E’ key. Once an item is in the inventory, it can be selected by pressing the left mouse button once. You can tell when an item has been selected as it will go red in colour. Items can also be selected by pressing the number key that corresponds to their slot in the inventory. Hovering over an item with the cursor will reveal extra information regarding that item. Once an item has been selected, it can be used to solve a puzzle. This can normally be done with the ‘E’ key again, but with pressure plate puzzles the player must instead press ‘C’ to drop an item from the inventory. With regards to corrupted items, these items require specific inventory slots to be open in order to be collected. For example, in level 3, the corrupted statue can only be picked up in safe mode when the 1st inventory slot isn’t full. Other items do not have requirements such as this, and will fill any inventory slot when picked up.

* Dialogue System

Dialogue is automatically triggered by the player when they reach different points in gameplay. These points include the loading up of a new level, interacting with an object, when the virus makes a move, or when a door/gate opens. The dialogue appears as text in a white rectangular box on the bottom of the screen. An icon will be showcased to the left of the text box to indicate which character (protagonist or Virus) is talking. The dialogue box appears and disappears on screen with a small pop or jump animation.
The text within the dialogue box runs on a coroutine to display letter by letter. Players can skip this coroutine by pressing the left mouse button once. Pressing the button again will progress the dialogue to the next line. Once all dialogue lines have been exhausted, the dialogue box will disappear again.
This system is crucial to gameplay as it acts as tutorialization for the player. When new mechanics are introduced, or when key narrative beats occur, dialogue guides the player and explains what happened to them. Additionally, The Virus’s property as a computer virus would not be nearly as apparent without the dialogue system. This system makes The Virus’s moves feel intentional rather than arbitrary.

Gameplay Actions and Challenges

Allowed Actions

- **2D Grid Movement:** The player is allowed to move on a grid with WASD or Arrow Keys.
- **Map Switching**: The player is allowed to switch from the normal map to the corrupted map with * the ‘Q’ key.
- **Inventory:** * The inventory is always present during gameplay, but objects in it can be selected with the left mouse button or associated numbers (1 - 4). Items can also be dragged into available slots by holding the left mouse button down and moving the cursor.
- **Interact:** The player can *collect items by pressing the ‘E’ key. These can be used to interact with locks or scanners using the same ‘E’ key.
- **Dropping items:** Items can be dropped when selected in the inventory, and then pressing the ‘C’ key.
- **Pausing/Restarting:** A game menu can be brought up by pressing the ‘esc’ key. This brings up a menu option which will automatically pause gameplay. An option is available to also restart a level by pressing the labelled ‘restart’ button.
- **Skipping Dialogue:** Dialogue is played automatically with letters revealing themselves individually. To show the full dialogue line, you can press the left mouse button once to complete the dialogue line and skip the letter revealing. You can press the left mouse button again to progress to the next line of dialogue.

Challenges

Puzzle Interactions

- ***“Key-to-door” Puzzles:*** Doors which require keys to open. These doors either lead into new rooms or propel the player to the next game level.
- ***Pressure Plates and Statue puzzles:*** Square plates that open a designated gate when an appropriate statue has been dropped on it by the player
- ***Inventory logic puzzles:*** These are puzzles that feature the player having to read item description on item they pick up to find clues or hints to solve a puzzle. These are are also puzzle where the player has to rearrange the items in their inventory to make space for more items to be picked up between modes
- ***Inventory scanner puzzles:*** Scanners which inspect the player’s inventory for specific items. If the player has all the items required to enter the room, the door slides open. Items required for scanners are not consumed and removed from the inventory upon opening a scanner door. This incentivises the player to inspect their maximum inventory capacity and make adjustments where needed.
- ***Impossible time-based puzzle sections (control panels):*** Control panels that the player manually activates to start a timer for a puzzle specific puzzle section. When the player finds the solution of that puzzle section, the timer is canceled and the player can

proceed forward. Upon failure, the puzzle section will reset all items the player has collected, resulting in the player having to restart the puzzle section.

Virus Influences

- ***Repositioning Items***: The virus can reposition items into different item cells intently or upon being triggered.
- ***Item corruption:*** The virus can corrupt items, making them unable to be used, picked up or dropped in normal mode.
- ***Grid cell blocking:*** The virus can block grid cells, making the map inaccessible from those grid cell points.

Normal Mode Limitations

- ***Invalid Corruption item pickups:*** Corrupt items are unable to be picked up and used in normal mode.

Safe Mode Limitations

- ***Single Available inventory slot:*** The player cannot only use/interact with items in their first slot. All other items are blocked from being selected, used or dropped.
- ***Pickup/Drop limitation:*** Only corrupt items can be picked up and dropped in safe mode. All other items require the player to switch to normal mode for pickups.
- ***Inventory Rearrangement limitation:*** The player only having a safe mode inventory capacity of 1 item means that items cannot be rearranged in safe mode and are blocked from being moved.
- ***Door interactions:*** To further incentivise the player to switch back to normal mode, the player cannot open/ interact with doors in safe mode. This restricts the player from completing the level in safe mode.

Design Process

Milestone 1 Iterative Process

Coding Process

General Process

When making the game different problems need to be solved to make a game which the group wants to make. These problems then get solved with the code, where the code is written in a way that follows the rules set out in the project plan document.
If new problems get found during the development of the game the code either gets changed or new scripts get added to solve the problem.

Code inheritance

With some of the code in the project it shared a lot of similarities with other parts of the code, an example of this is the items and interactions found in the game, these scripts (children scripts) were then made to inherit from a parent script. The children script can be used with existing scripts that use the parent script without needing to allow the existing scripts to change their code each time a new interaction or item is made.

Code improvement

During the process of coding the first solution is usually not always the best solution. The code gets looked over and then evaluated based on:

- The complexity of the solution
    - This is if the code uses more functions than necessary or if it gets to the solution in a roundabout way.
- The run time of the solution
    - Does the solution take longer than usual to run? If it does, does it actively affect the player by making the game slower? If not then the solution can be kept and if it does then the solution needs to be redone.
- The compliance of the solution
    - Does the solution comply with the rules set out for coding in the Project Plan Document.

If the solution does not succeed in these evaluations then the code gets redone to be successful with these criteria.

Level and Design Process

This section documents the logic and design choices behind the implementation of the levels within the game. The section primarily focuses on the thinking, structure and puzzle choices behind each level rather than a technical description of the level building process. **Any extensively detailed explanations of implementation and technical design of the level is documented in Phemelo Tsogang's (2543064) individual design document**.

Level 1 Implementation

The core focus of this level was to serve as a tutorial for the player to learn the base mechanics they can use to solve puzzles in the game. This included movement, interacting with objects, picking up and dropping objects, selecting objects in their inventory and using these objects in the player's inventory. Since puzzles in each level will have different mechanics with different object interactions, this first level was used to purely explain the consistent mechanics and systems the player can use to all puzzles within the entire game. The design choice was then obvious to leave the Virus changes and interferences out of this level, as this would confuse players and seem like glitchy behaviour while players are still grasping their own capabilities.

The puzzles in this level are relatively simple but logic-based by design. Each room in a level is designed to make the player use inventory logic to solve the puzzle. The most direct example of this in Level 1 is a puzzle where the player finds 2 pieces of paper with 2 different codes on them. Both codes are the password to the door but the solution to the puzzle is the one paper

indicates that the password it contains is an obsolete code from 6 years ago. Therefore, the piece of paper that the player must select and use to interact with the door is the more recent paper.

Level 1 is filled with similar logic based puzzles all designed in an effort to familiarize the player with movement, interacting and inventory usage through repetition. The inventory was also designed to always be on the player's screen so that the player always knows what is in their inventory and how many items are in their inventory.

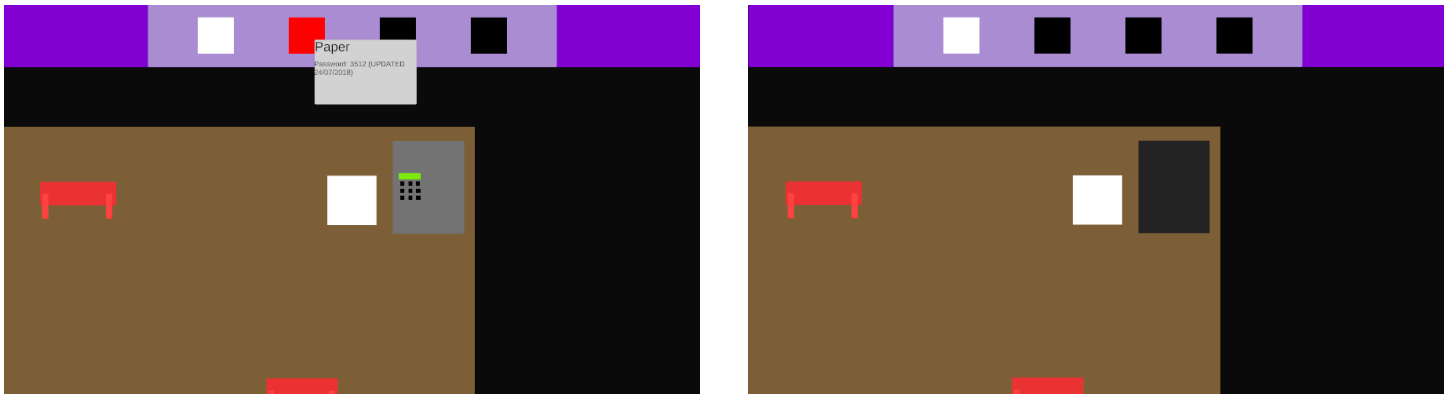


Figure 6: Player selecting paper with updated password which unlocks door.

Level 2 Implementation

The main focus of level 2 was to introduce the antagonistic Virus and its mechanics, the switching between safe mode and normal mode, as well as the interaction between these two systems.

The thought process behind this implementation was to ramp up the difficulty within the puzzles through the puzzles themselves, as well as the interferences caused by the virus. The Virus and its interferences are introduced in a small “key-and-door” puzzle section which the player has previously seen. It was important that this puzzle section was something the player had already seen to create an expectation in the player on how they would solve the puzzle (open the door with a key). This expectation is then subverted when the Virus is introduced which blocks cells between the player and the key. The player then has to discover and switch to safe mode to get past the blocked cells and reach the key. This brief interaction shows the Virus making a change to the puzzle which subverts the player's expectation, which then prompts the player to use mode switching to solve that change. This directly correlates with our hypothesis of the virus subverting player expectations while simultaneously showing the relation and connection between the 2 systems we cloned.

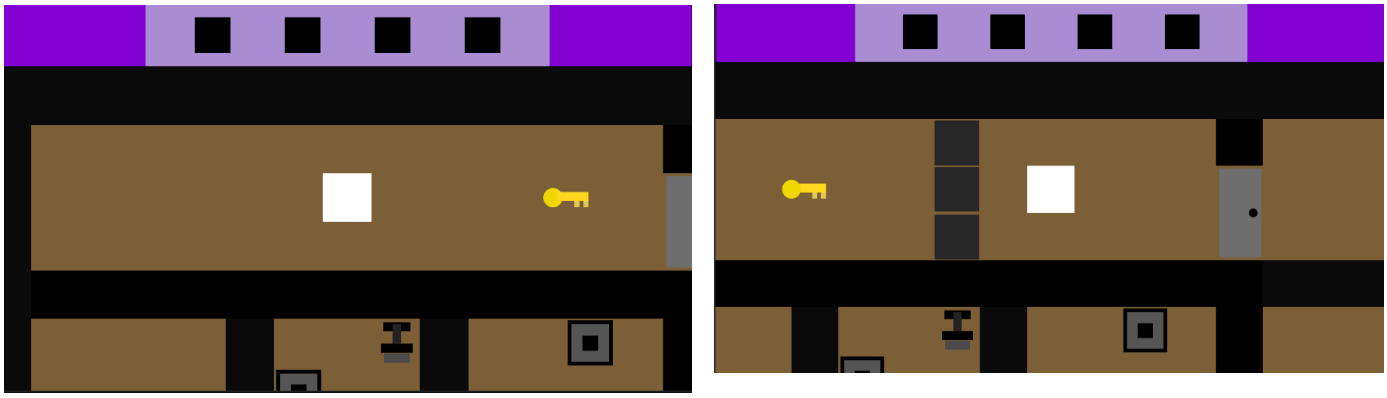


Figure 7: Left Image - Player trying to open key to open door puzzle section. Right Image - The virus switches the position of the key and blocks cells for the player.

This type of interaction is then scaled to a higher degree with the next puzzle section which has the player trying to solve a pressure plate puzzle with the Virus blocking cells and switch pressure plates around so that certain gates are closed, thus subverting the player's expectation of the puzzle solution.

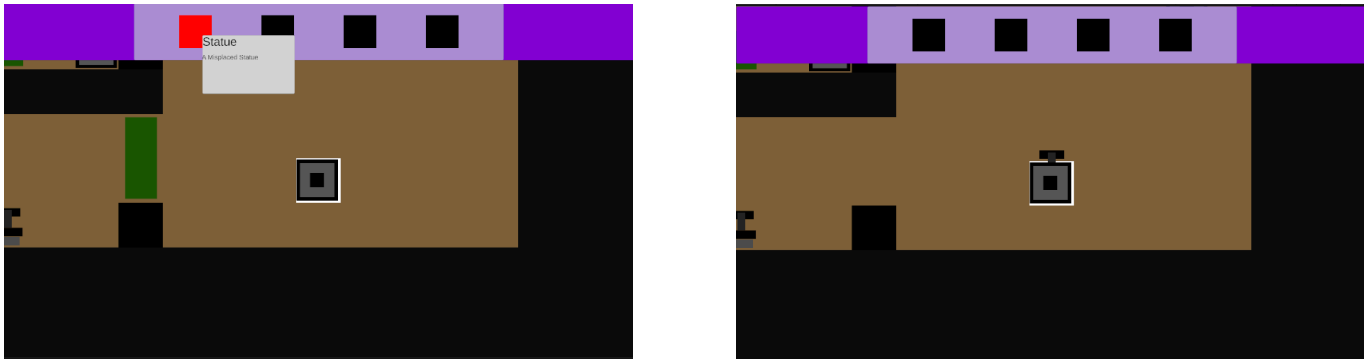


Figure 8: Pressure plate puzzle.

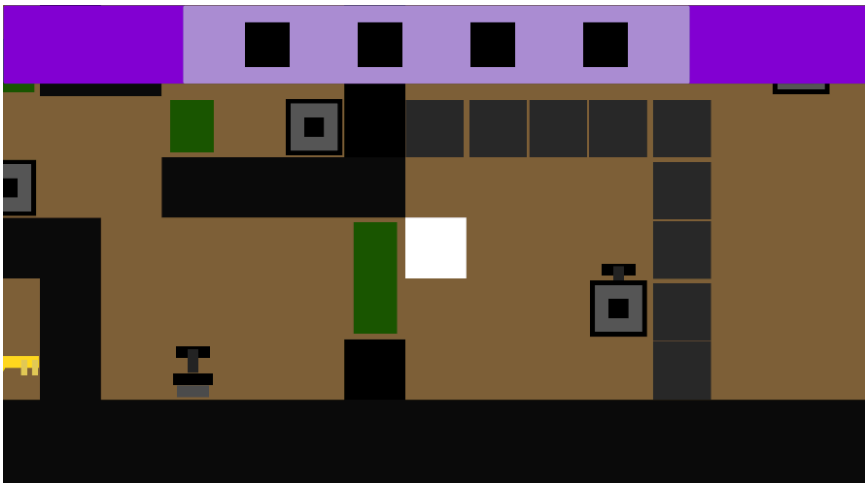


Figure 9: Virus switching pressure plates and blocking cells once the player has solved the puzzle in "Figure 9".

Overall, this level sets up the and familiarizes the player with the exchanges and interactions that they will have with puzzles and The Virus across the entire game.

Narrative Process

Starting the narrative, we had the core idea that the player would act as a game designer who is playtesting their game. During playtesting, they discover a Virus in their code, which begins to manipulate gameplay as you, the player/main protagonist, work to eradicate this Virus.

From there, we have built the narrative into a dialogue script. As you play the game, the protagonist begins to speak. As you are the protagonist, you must think of this as you talking to yourself. What is said will appear on the screen in a dialogue window. This use of dialogue also allows The Virus's manipulation of play and its reasoning to be more apparent. Using dialogue has made the game feel more alive, as you feel more immersed in the overall narrative. Additionally, we have used the protagonist's voice as part of the tutorial. You are "checking" each mechanic works, so this explains to the player how things in-game work.

This narrative dialogue, or dialogue script, saw two major iterations. The initial iteration, whilst functional, lacked depth of engagement with the game. For example, The Virus would move a key's location, but the dialogue did not incorporate any verbal reaction from the protagonist. This lack of feedback was not useful for game-feel, and so it needed to be worked on. We need players to understand that every "glitch" in the game is intentional, so dialogue must reflect this. Alongside this, the initial iteration saw The Virus having a friendly and very humanistic voice. We decided to change this as it felt too much like a hacker or other human inside the game, which works against our narrative having a virus inside it. The Virus needed to be more formal, technical and computational. Our second iteration of the dialogue script worked on this feedback and was ultimately the one that was incorporated into the game for Milestone 1.

THE VIRUS

Wait, how'd you do that? You just disappeared for a second and then BAM, you got the key.

I thought I had you fooled.

PROTAGONIST

AHHH, WHO ARE YOU?

THE VIRUS

Oh, me? I'm no one or no thing of importance. I'm just here to keep you company :)

Figure 10: the initial meeting interaction between The Virus And protagonist in the script Iteration 1.

THE VIRUS

[NEW ENTITY OBSERVED]

Co-existence... plausible? Calculating percentage of successful co-existence...

...76%

PROTAGONIST

AHHH!

WHAT is going on?!

THE VIRUS

Scanning database and interweb to identify human response...

90% chance emotion is fear.

Figure 11: The same interaction as figure 10, but reworked
To reflect feedback for Iteration 2.

* Milestone 2 Iterative Process

Coding Process

General Process

With the adding of new scripts and mechanics the primary focus of the coding was getting the scripts to work together. To do this some scripts needed to be refined by having their output easily be used by other scripts. When referring to old scripts I refer to scripts that were done before this milestone.

Additions to old scripts

Old scripts sometimes needed other elements in the scene in order to fulfill new functions. When this happens it can mess up the scenes that use these scripts, thus it needs to be coded in a way where the new addition does not cause errors in the scene. The easiest way to do this is just to add null checks with the new addition. This prevents the script from giving an error when the code executes the new function in the old scene.
In more severe cases where the output of an old script needs to change, that would cause all other scripts that depend on the specific old script to also change. This situation has not happened yet, but if it happens we plan to deal with each script individually.
The most common addition was just an output addition on specific functions to the old scripts. This was usually events where the event gets invoked in the function. This way of output helped the scripts that used the events get their input, while not changing anything that the scripts dependent on the old scripts relied on.

Script coding process

To prevent the severe case situation where each script needs to be changed when a script's output changes the script needs to be coded in a certain way. This way is that the scripts' output needs to have null checks. This means that the script will still work and give an output even if no script explicitly uses that output. This allows the scripts that use that output to only need to check for an output from the input script. This allows us to be able to freely add outputs to the input scripts without affecting the scripts that are dependent on that output. If we continue to follow this way of coding then we will prevent situations where multiple scripts need to be modified in order to accommodate changes.

Future iteration process

For this submission all the code should already be refined to a point where new outputs can easily be added to existing scripts. If this is not the case then the following should be done

1. Check what the existing code outputs
2. Check if the code output can be changed
3. Check whether the scripts that use the output would also need to be changed
4. Change the code if needed.

If the previous steps are followed then the decision needs to be made if the code can be changed or if the new addition needs to be discarded and a new solution needs to be thought of.

Level and Design Process

Level 3 Implementation

The primary focus of Level 3 was to fully flesh out the capabilities of The Virus - namely, the Virus' ability to move item positions, corrupt items and block map grid cells. This was done for two reasons. The first reason was to create a more complex level where virus bombardment creates new puzzles that the player has to figure out and adapt to. The second reason was to connect the 2 main systems of the game more deeply, to show that the antagonistic virus and map switching work in tandem and not as two separate unrelated gameplay systems. This second reason for implementation is a direct iteration from an issue identified in Milestone 1, where the group generally felt that Safe Mode was being used as the solution to all puzzles instead of as "one half" of a solution which needs to be solved using BOTH safe and normal mode.

To implement this sufficiently, the map size needed to be expanded to fit a naturally and progressively more difficult overall puzzle area than level 2. Therefore, it was decided that Level 3 would feature 3 parts: a small introductory puzzle section expressing a new puzzle and Virus mechanic, a second puzzle section using the learned puzzle mechanic with moderate Virus influence, then a larger final puzzle section where the new puzzle mechanic is used with the rest of the players knowledge to create a complex puzzle solution. In this third part, the Virus is fully active and creates situations where the player has to utilize every single mode switching advantage and disadvantage, as well as every learned puzzle interaction to enter the final door leading to the next level. This includes "key-to-door" interactions, pressure plates, inventory-logic and management puzzles, as well the newly added inventory scanner puzzle interaction.

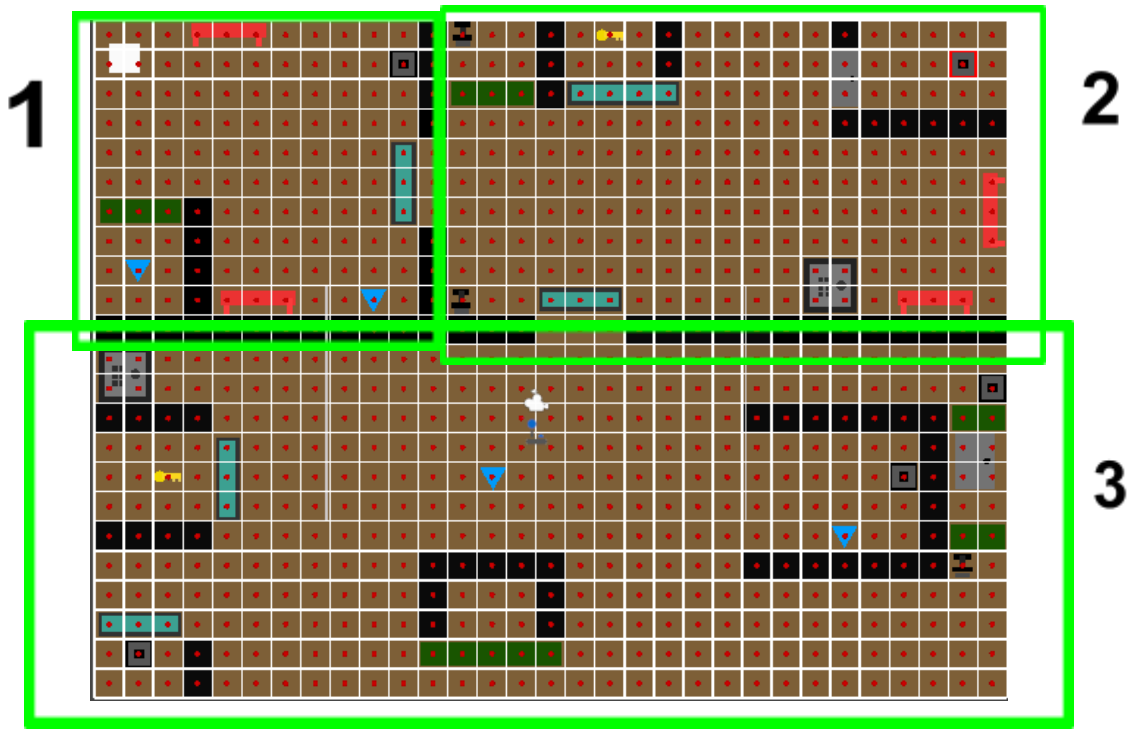


Figure 12: Larger Level 3 map grid with 3 puzzle sections parts.

The inventory scanner puzzle interaction in particular was used to create a situation where the player has their inventory at maximum capacity and has to manage it. This fully fleshes out the map switching advantages and disadvantages of both modes as there are situations requiring the player to use all 4 of their inventory slots while safe mode restricts them to 1 available inventory slot. The player therefore has to keep switching between modes, rearranging their inventory and planning out their inventory before switching between modes to adapt to puzzle requirements or Virus influence. Explicit puzzle examples are highlighted in Phemelo Tsogang's Individual Documentation.



Figure 13: Filled Inventory with blocked slots in Safe mode.

Along with this constant inventory management, an extra layer of difficulty is added with corrupt items. These items cannot be picked up or interacted with in normal mode as they have been corrupted by The Virus and thus forces the player to switch to safe mode. The complexity in this implementation is the player only being able to use their first inventory slot in safe mode and must thus empty out their first inventory slot in safe mode to accommodate for this safe mode disadvantage. Thai ultimately makes the player have to consider both modes more deeply and

thus ensures that the player is not just arbitrarily switching between modes without thinking through how switching will have a cause an effect relationship between the puzzle state and inventory system. This is ultimately the complexity the group wanted to reach with mode switching, whereby the player actually has to understand the consequences and advantages of switching without just thinking that “safe mode is the way out”.

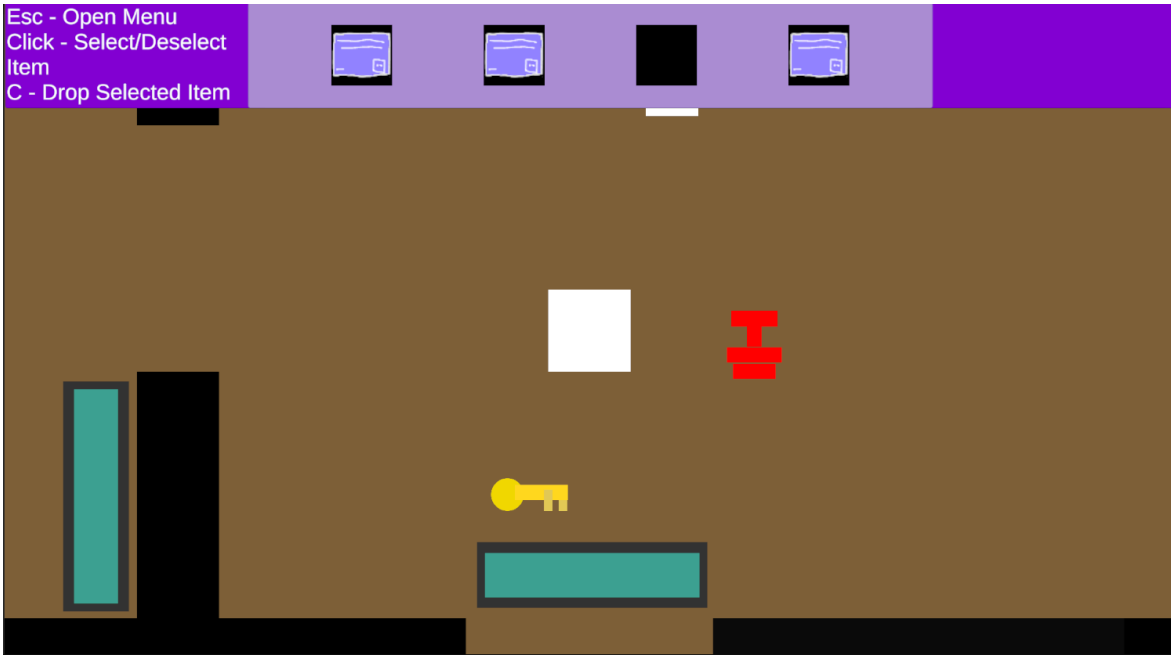


Figure 14: Virus item corruption (red statue)

Level 4 Implementation

Level 4 is the final planned level for the project as a whole and thus needed all mechanics to be complete. The previous 3 levels have established how the player has to be creative in their adaptation against the virus in order to find a puzzle solution. To test this at its most extreme expression, the *timer control panels* were implemented. These control panels add a seemingly impossible time-based challenge to the player to solve the puzzle. The purpose of the panels was to create a situation where the player is literally unable to complete the puzzle within the time limit allowed. Thai creates a situation where the player has to test how different systems in the game affect the timer. The player would then use the pause menu, where the player is still able to move but is unable to interact with puzzle objects. The timer does not run down in the pause screen and therefore the player is incentivised to use this menu to progress through the level without losing any of their remaining time.

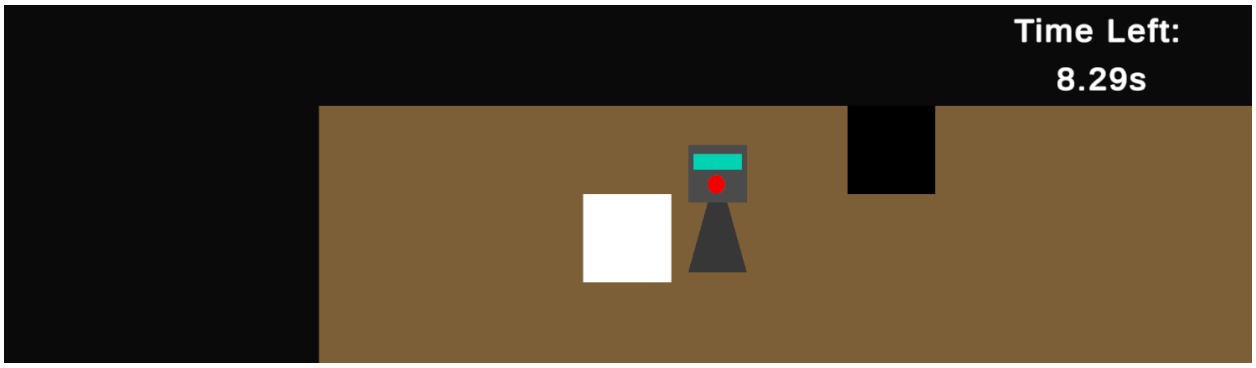


Figure 15: Control Panel initiating time-based puzzle

This puzzle mechanic was designed with the intention of showing how the virus can not only affect objects in the game space, but mechanics themselves when fully aggravated. This puzzle mechanic was also used to demonstrate how a “known bug”in one iteration of a game can actually be turned into a working feature in an updated iteration. Currently the level is in the prototyping phase, with the pause menu mechanic being playtested to keep the game balanced. The prototype is meant to finish off the core gameplay loop of the entire game by showing an extension of the Virus’ abilities. By the end of the development of this mechanic, the interactions lists for the game were completed, allowing for narrative to be developed with and around their interaction rules.

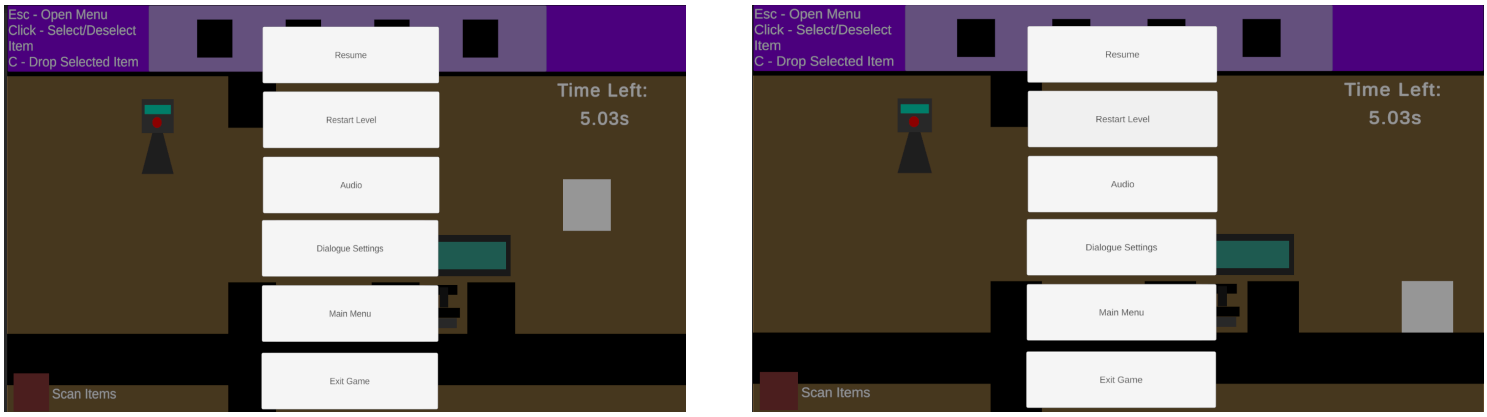


Figure 16: Pause game mechanic showing player moving through grid map without using any remaining time.

Narrative Process

The beginning of this process saw me going back into my dialogue for level 2, and changing a couple words around. This wording change was necessary to improve clarity in the instructions given to the player.

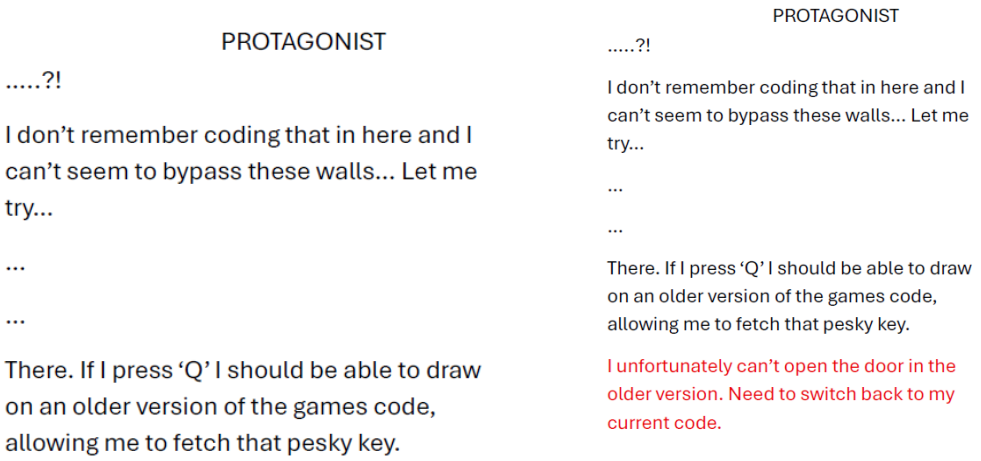


Figure 17: Implementing new dialogue lines to improve clarity (before on left vs after on right).

Alongside this, I also changed some of the language I had used for milestone 1. Specifically, the use of the word "backdate". Originally, I explained the concept of safe mode as an older version of the game, and so to remove The Virus the playtester was going to backdate the game. I decided to move in the direction of the playtester instead using old "code" of the game as it felt more realistic to the real iterative journey game designers face. You change one element of code, rather than the entire version of the game when going back to the older version. Additionally, The Virus lives inside the playtesters code, so by the playtester talking about old code, it links better to the discovery of The Virus.

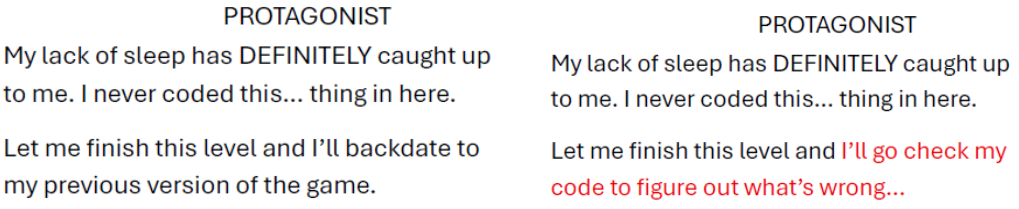


Figure 18: Old (left) vs new (right) lines of dialogue

After these changes were made, I continued writing dialogue as planned. I wrote about the playtesters' discovery of The Virus inside their code, and the related emotions. I also added in a reflective piece about the playtesters keycards in their real life office to help characterize the playtester and create a bit of humor. Moments like these also help the player feel connection to the game's storyline as they feel more invested in the characters by knowing more about their lives.

Playtesting Build 1

General

Playtesting was done with the milestone 1 build. The milestone 1 build included the basic systems of the game with little communication design. The levels included are level 1 and level 2.
The mechanics present are:

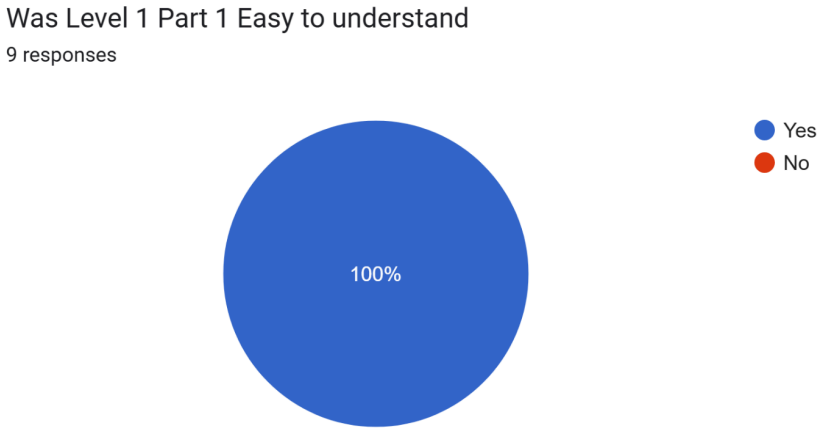
- Moving
- Interacting (Objects and items)
- Selecting/Deselecting items
- Switching modes (Safe -> Normal or Normal -> Safe)
- Dropping items

The main parts of the game that were being playtested were:

- The flow of the levels.
- The speed of the player
- The complexity of the puzzles
- The tutorialization and
- Narrative

Flow of the levels

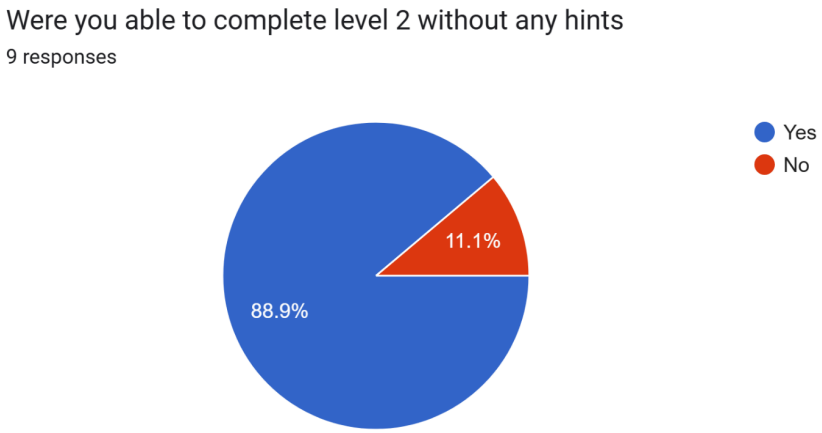
The flow of the levels was tested by seeing how the play testers played the level. We were then looking at if the play testers played the levels the way we wanted the levels to be played or if something was not communicated properly and the playtesters got stuck. These were the results.



All 9 playtesters could understand Level 1 Part 1.



All but 1 play tester understood what they learnt in level 1 part 1. The reason the last playtester did not fully understand is because the narrative explained the controls while that playtester skipped the dialog. This means that we need to have communication of the mechanics given to the player through sources other than the dialog.

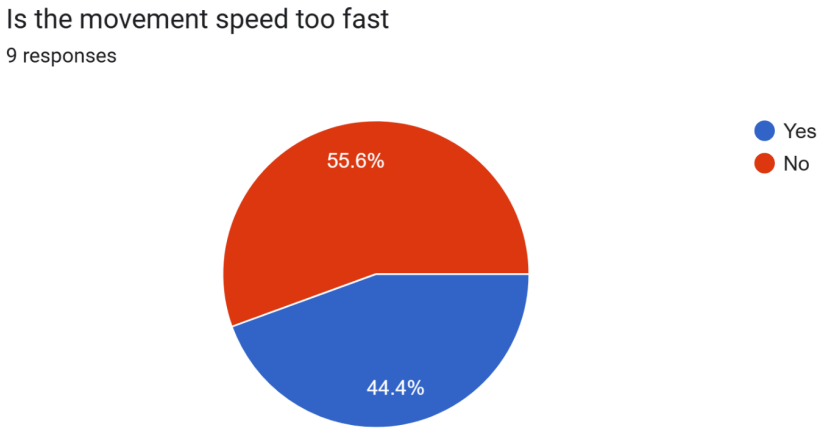


All but 1 of the playtesters could complete level 2 without hints. The reason given by the 1 playtester was that they forgot the drop button. This indicates that our communication of the keybinds may not have been clear enough.

With the evidence provided it shows that the flow of our levels were interpreted the way we wanted them to be interpreted, thus we do not have to change the general flow of the levels.

The Speed of the Player

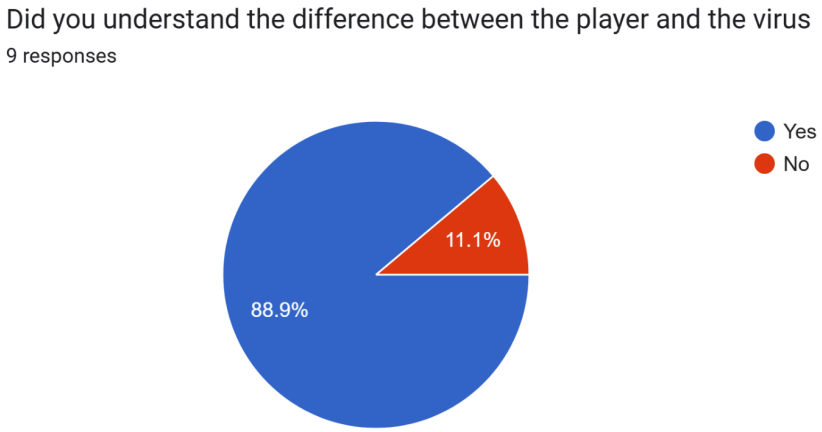
During the playtesting we also asked if the player character moved too fast. This is because we set a speed for the player character which was fast, but we liked this speed and kept it with the goal in mind that it will be changed if the playtesting shows tha the speed is too fast.
We got the following results.



With the results we can see that the speed of the player is too fast for 4 out of the 9 playtesters. This caused us to change the speed of the player to a slower speed.

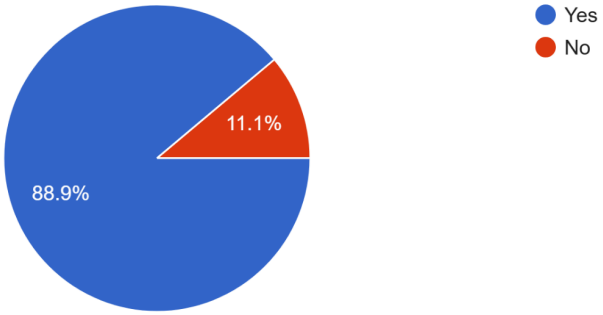
Narrative

The main focus of the narrative playtesting is to know if the player can differentiate between the player (protagonist) and the virus (antagonist). With that in mind these were the results of playtesting.



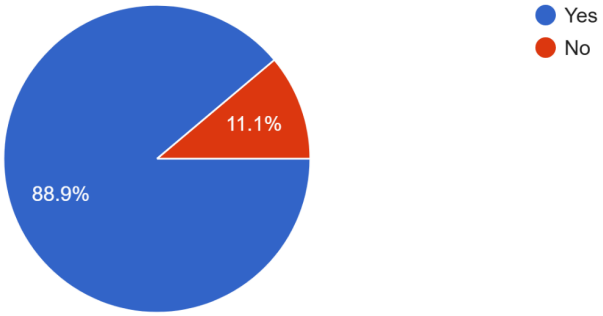
Did you understand what the virus changed in Level 2

9 responses



Did you understand what to do when the virus changed the puzzle in Level 2

9 responses



With the responses shown above we can conclude that the narrative of identifying the protagonist and antagonist is effective in the game. This shows us that the narrative we have set out to show is on track.

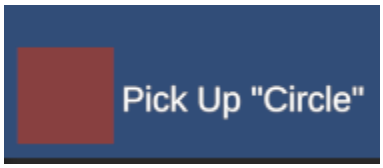
Change brought from feedback

With the feedback that we got from the player speed we changed the value of allowing the player to move every 0.5 seconds up 1 second. This slowed down the speed and stopped the player character from sometimes jumping over a grid space that they wanted to move to.

The feedback about the communication design that we got reinforced our ideas of what communication design we were doing for milestone 2. These are:

- Indicating what the player can interact with
- Indicating when an interaction fails.
- Giving small reminders about the controls.

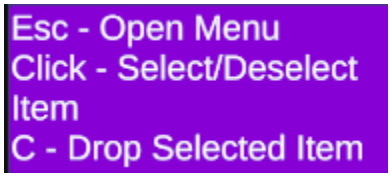
Indicating what the player can interact with is shown on the bottom left of the screen by a temporary red box and some text.



The red box will be replaced by an image of the interact key.
Indicating when an interaction fails is done by floating text.



Giving a small reminder about the controls is temporarily done through the inventory background.



Known Bugs

Bug	Description	Status	Solution
Movement Clipping bug	The player character can use the speed of moving diagonally in the grid to sometimes clip into objects.	Fixed	The raycasting of the player was not long enough to check diagonally thus it was extended
Moving while game paused	The player can move while the game is paused. The group is looking at possibly implementing this bug as a puzzle.	Feature	The group decided to implement the bug as a feature for time based puzzles on level 4
Dialog while game is paused	Dialog can still happen and be skipped when the	Not Fixed	Not yet implemented

	game is paused		
Virus effects in Safe Mode	If the player collides with a virus trigger in safe mode then normal mode acts as safe mode for that virus change	Partially fixed	Dialog of the virus can still trigger in safe mode, thus the group needs to be careful when setting dialog triggers.
Scanner panel stays forever	If the player goes into safe mode the scanner panel stays on the players screen.	Not Fixed	Not yet implemented

Bibliography

Images

Figure 1: Allred, A. (2012). *The Silent Age: A Reflective Critical Review.* Medium. Available at: https://medium.com/@avaallred03/the-silent-age-a-reflective-critical-review-2500d654e26

Figure 2: The Walkthrough King (2021) *There Is No Game: Wrong Dimension*. Available at: https://www.walkthroughking.com/text/thereisnogamewd.aspx (Accessed: 30 August 2026).

General and Research References

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